Mahmoud Alyosify

AI Instructor & Technical Trainer | Curriculum Design, Applied ML Teaching & Technical Content
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PROFESSIONAL SUMMARY

AI Instructor and Technical Trainer with **6+ years** teaching computer science and machine learning to **10,000+ learners** across **700K+ YouTube views** — and, crucially, a practitioner who still builds. Part-time instructor at Egypt's **National Telecommunication Institute (NTI)** since 2025 (**Machine Learning** and **AI for Business**), and authored Arabic-language curricula in **discrete mathematics, algorithms, probability, statistics and graph data mining** for a region under-served by technical content in its own language. Currently completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship** and working as an **AI Engineer Intern at Injaz Tech** on LLM systems, which keeps teaching material grounded in current practice. **1st Place, Science & Technology Content Competition**.

TEACHING & TRAINING EXPERIENCE

Part-Time Instructor — Machine Learning & AI for Business *Jul 2025 – Present*
National Telecommunication Institute (NTI) Egypt

- Teach the **Machine Learning** and **AI for Business** modules of NTI's Summer Internships Program to engineering and computer-science trainees, on separate part-time engagements.
- Machine Learning: 90-hour hands-on bootcamps covering regression, SVM, KNN, Naive Bayes, PCA and neural networks through Python labs.
- Write final exams and assess end-to-end capstone projects.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built and taught Arabic-language technical curricula in **discrete mathematics, algorithms, probability, statistics** and **graph data mining**, reaching **10,000+ learners** and **700K+ YouTube views**.
- Authored courses end to end — syllabus design, lesson scripting, recording, editing and ongoing learner support — sustaining high ratings across cohorts.
- Addressed a genuine gap: rigorous CS fundamentals taught in Arabic, for students who lose material to the language barrier before they lose it to difficulty.

Ambassador *During undergraduate study*
Information Technology Institute (ITI), Assiut Assiut, Egypt

- Represented ITI on campus, promoting technical training pathways and guiding students toward relevant programs.

EDUCATION

MSc in Artificial Intelligence *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) *Sep 2019 – Jul 2023*
Assiut University, Egypt GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TEACHING & TECHNICAL SKILLS

Teaching & Delivery	Curriculum & Syllabus Design, Hands-On Lab Development, Exam & Assessment Design, Capstone Project Supervision, Lecture Delivery, Online Course Production, Technical Content Creation, Video Production & Editing, Learner Support, Public Speaking, Bilingual (Arabic / English) Instruction
Subjects Taught	Machine Learning, Linear & Logistic Regression, SVM, KNN, Naive Bayes, Neural Networks, PCA & Dimensionality Reduction, Supervised Learning, Discrete Mathematics, Algorithms, Probability & Statistics, Graph Data Mining, Programming Fundamentals
AI & ML	PyTorch, TensorFlow, Scikit-learn, Hugging Face, Large Language Models, Fine-Tuning (LoRA, QLoRA), RAG (LangChain, LangGraph), Computer Vision, Reinforcement Learning (PPO, DQN), Multi-Agent Systems
Engineering	Python, SQL, C++, C#, C, JavaScript, R, MATLAB, AWS, Docker, FastAPI, Streamlit, ONNX, Git/GitHub, Linux

PRACTITIONER WORK BEHIND THE TEACHING

- **MinimalLM** — MSc research on training-free decoding-time control of LLM verbosity: a **25.5–34.9%** token cut proved to be truncation, so an NLI-licensed elision method was built that beats truncation on every model.
- **HORUS Sentinel** — autonomous multi-agent OSINT platform with a knowledge graph and self-hosted fine-tuned LLM.
- **SimCLR-Vision-SSL** — self-supervised vision pipeline, **42-experiment** augmentation sweep, **84.30% top-1**, ONNX/FAISS deployment.
- **RL-Cloud-Autoscaler** — custom Gymnasium environment with PPO and DQN agents optimising a cost–latency trade-off.
- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** records, deployed on AWS.
- **Also:** Automated Multimodal Agent · SCRAP · AudioShield · Vitalism Solution (rPPG) · Human vs. Agentic Pull Requests.

ADDITIONAL PROFESSIONAL EXPERIENCE

AI Engineer Intern *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) Maadi, Cairo, Egypt

- Building an agentic RAG assistant (LangChain, LangGraph, FAISS, FastAPI) and documenting each milestone in study guides and weekly presentations.

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

AWARDS, CERTIFICATIONS & LANGUAGES

1st Place — Science & Technology Content Competition

Digilians Presidential Scholarship — Queen's University

Machine Learning Specialization — DeepLearning.AI

HCIP – AI (Huawei)

Ideal Student Award — Assiut University, 2022–2023

1st Place — Smart Cities Hackathon 2022

AWS Academy — ML Foundations, ML for NLP, Data Engineering

Communication Skills: Star Presenter on Panels

- **Languages:** Arabic (Native) · English (Professional Working Proficiency).